



YORDAMCHAI

AI-Powered Ta'lim Platformasi

MASTER PLATFORM BLUEPRINT

Enterprise Audit & Architecture Documentation · v2.0

Versiya	2.0.0
Sana	2026-06-25
Muallif	CTO Office
Holat	KONFIDENSIAL
Til	O'zbek (Latin)

1. EXECUTIVE SUMMARY

YordamchiAI — React 19 + Supabase + Gemini 2.5 Flash asosida qurilgan AI-powered ta'lim platformasi. Hozirgi holat: **MVP bosqichi (5.3/10)**. Asosiy kuchli tomon — AI chat va Supabase arxitekturasini. Asosiy zaif tomon — test yo'qligi, streaming simulyatsiya, accessibility nol.

1.1 Platform Umumiy Bahosi

6.0

Arxitektura

7.0

Database

7.5

UI Sifati

5.5

UX Sifati

7.5

AI Tizimi

6.0

Xavfsizlik

4.0

Performance

5.0

Skalabillik

1.0

Test Qamrovi

5.3 / 10

UMUMIY BAHO — MVP bosqichi, production-ready emas

6 TA KRITIK MUAMMO PRODUCTION'GA CHIQISHNI TO'XTATMOQDA:

(1) Role self-selection: har kim admin bo'la oladi | (2) avatars bucket yo'q: avatar upload 404 | (3) Error boundary yo'q | (4) Code splitting yo'q: 1.37MB | (5) ProfileContext ishlatilmayapti | (6) Email verification o'chirilgan

2. PLATFORM MODULLARI

Modul	Holat	Tayyor %	Asosiy Muammo
Autentifikatsiya	Qisman	75%	Role bug kritik, email verify yo'q, OAuth yo'q
Talaba Paneli	Qisman	50%	Dashboard static, profile eski, homework yo'q
O'qituvchi Paneli	Qisman	40%	Progress view yo'q, analytics yo'q
Admin Paneli	Qisman	50%	Analytics yo'q, audit log yo'q
AI Assistant	Qisman	80%	Streaming simulyatsiya, rate limit yo'q
Darslar	Qisman	55%	Video player yo'q, progress yo'q
Davomat	Qisman	55%	Calendar yo'q, export yo'q
Testlar	Qisman	55%	Timer noaniq, natija detail yo'q
Yutuqlar	Qisman	60%	Sertifikat DB yo'q
Sertifikatlar	Yo'q	5%	Faqat client-side PDF. DB jadval yo'q!
Foydalanuvchi Profil	Kritik	35%	ProfileContext qurilgan, lekin ISHLATILMAYDI
Xabarnomalar	Yo'q	0%	DB jadval yo'q, UI yo'q
Uy Vazifalari	Yo'q	0%	Hech narsa yo'q — jadval ham yo'q
Leaderboard	Yo'q	0%	UI yo'q
To'lovlar	Yo'q	0%	Hech narsa yo'q
Qidiruv	Yo'q	0%	Global search yo'q
Lokalizatsiya	Qisman	70%	uz/ru/en bor, ba'zi string hardcoded
Supabase	To'liq	90%	pgvector yo'q
White Label	Yo'q	2%	Hech narsasi konfiguratsiyadan kelmaydi

3. DATABASE AUDITI

3.1 Jadvallar (16 ta)

Jadval	RLS	Holat	Muammo
profiles	✓	●	updated_at yo'q, username yo'q, preferences yo'q
subjects	✓	✓	Yaxshi tuzilgan
groups	✓	●	deleted_at yo'q
student_groups	✓	✓	Composite PK to'g'ri
teacher_subjects	✓	●	UI'da kam ishlatiladi
lessons	✓	●	order_index yo'q, deleted_at yo'q
lesson_attachments	✓	✓	Yaxshi
settings	✓	●	Runtime'da o'qilmaydi!
attendance	✓	✓	Unique constraint to'g'ri
tests	✓	●	JSONB questions DB'da tekshirilmaydi
test_results	✓	●	JSONB answers tekshirilmaydi
achievement_definitions	✓	●	Qisman
user_achievements	✓	●	Qisman
user_score_snapshots	✓	●	Qisman
ai_conversations	✓	✓	Trigger mavjud
ai_messages	✓	●	token_count yo'q!

3.2 ER Diagrammasi

```
profiles (id)
├──[1:N]── student_groups → groups → subjects
│
└── profiles (teacher_id)
├──[M:N]── teacher_subjects → subjects
├──[1:N]── attendance → groups
├──[1:N]── test_results → tests → groups/subjects
├──[1:N]── user_achievements → achievement_definitions
└──[1:N]── user_score_snapshots
```

```
└─[1:N]─ ai_conversations
      └─[1:N]─ ai_messages

lessons (group_id → groups)
  └─[1:N]─ lesson_attachments

[GLOBAL] settings (key-value)
```

● YETISHMAYOTGAN: homework, homework_submissions, certificates, notifications, announcements, audit_logs, payments, subscriptions, comments, leaderboard_snapshots — bu jadvallar YO'Q!

⚠ profiles.updated_at ustuni mavjud emas — bitta ALTER TABLE bilan qo'shiladi!

4. AI TIZIM AUDITI

4.1 Joriy Arxitektura

```
AIAssistantPage.tsx (700+ qator — MUAMMO)
├── EdgeFunctionProvider.complete()
│   └── supabase.functions.invoke("ai-chat")
│       ├── buildSystemPrompt(studentContext)
│       ├── Gemini 2.5 Flash API
│       │   ├── thinkingBudget: 0 ✅ tejamkor
│       │   └── parts.filter(!thought).join("") ✅
│       └── { response: text }
├── ai_messages (Supabase DB) ✅
└── StreamingMessage — SIMULYATSIYA ❌ (setTimeout batch)
```

4.2 Kuchli va Zaif Tomonlar

Xususiyat	Holat	Izoh
AIProvider interface	✅ A'lo	Model almashtirish uchun abstraction — strategik qaror
Student context injection	✅ Yaxshi	Guruh, dars, test, davomat AI'ga yuboriladi
Thinking parts filter	✅ Tuzatilgan	Gemini 2.5 Flash "thinking" partlari filtrlanadi
Haqiqiy streaming	❌ YO'Q	setTimeout simulyatsiya — texnik qarz
Rate limiting	❌ YO'Q	Bir user kvotani tugata oladi — moliyaviy xavf
Token tracking	❌ YO'Q	Xarajatni nazorat qilib bo'lmaydi
Context window	❌ YO'Q	Barcha tarix yuboriladi — token limit xavfi
AI memory	❌ YO'Q	pgvector kerak

5. TEXNIK QARZ

Muammo	Jiddiylilik	Ish hajmi
AIAssistantPage.tsx 700+ qator (SRP buzilishi)	Kritik	Katta
Avatar 3 parallel implementation	Kritik	O'rta
ProfileContext ishlatilmayapti	Kritik	O'rta
Hardcoded brand values (30+ fayl)	Yuqori	O'rta
Code splitting yo'q (1.37MB)	Kritik	O'rta
MockAIProvider bundleda	Yuqori	Kichik
Soft delete yo'q barcha jadvallarda	Yuqori	O'rta
settings jadval runtime'da o'qilmaydi	O'rta	O'rta
try_login.mjs va avatar-design.html repoda	Yuqori	Juda kichik
0 ta test (Playwright o'rnatilgan)	Kritik	Katta

6. TOP 50 MUHIM YAXSHILANISHLAR

#	Yaxshilanish	Muhimlik	Vaqt	Biznes ta'siri
1	Role security fix	Kritik	0.5 kun	Juda yuqori
2	avatars bucket yaratish	Kritik	0.5 kun	Yuqori
3	ErrorBoundary barcha layoutlarda	Kritik	1 kun	Juda yuqori
4	Email verification yoqish	Kritik	1 soat	Yuqori
5	ProfileContext → barcha sahifalar	Kritik	3–5 kun	Juda yuqori
6	Code splitting (React.lazy)	Kritik	2 kun	Yuqori
7	AI rate limiting	Kritik	2 kun	Kritik (moliya)
8	Token usage tracking	Kritik	1 kun	Kritik (moliya)
9	/reset-password route	Kritik	1 kun	O'rta
10	profiles.updated_at migration	Kritik	1 soat	O'rta
11	Haqiqiy Gemini streaming	Yuqori	5–8 kun	Juda yuqori
12	Toast notification system	Yuqori	2 kun	Yuqori
13	brand.ts config (White Label)	Yuqori	2 kun	Juda yuqori
14	AIAssistantPage → feature module	Yuqori	5–8 kun	Yuqori
15	AI context window sliding	Yuqori	3–5 kun	Yuqori
16	Homework moduli (DB + UI)	Yuqori	10–15 kun	Juda yuqori
17	Notifications tizimi	Yuqori	8–12 kun	Yuqori
18	Certificate DB + verification	Yuqori	5–8 kun	Yuqori
19	Leaderboard sahifasi	Yuqori	5–8 kun	Yuqori
20	Playwright test suite	Yuqori	8–12 kun	Yuqori
21	Soft delete barcha jadvallarda	Yuqori	2 kun	Yuqori
22	Teacher progress view	Yuqori	8–12 kun	Yuqori
23	Accessibility birinchi pass	Yuqori	10–15 kun	Yuqori (legal)
24	Avatar tizim unification	Yuqori	3–5 kun	O'rta

25	Google OAuth	O'rta	3–5 kun	Yuqori
26	Admin analytics dashboard	O'rta	15–20 kun	Yuqori
27	Teacher analytics	O'rta	8–12 kun	Yuqori
28	Design token system	O'rta	3–5 kun	O'rta
29	Feature-based arxitektura	O'rta	15–20 kun	Yuqori
30	Zustand state management	O'rta	5–8 kun	Yuqori
31	Video player integration	O'rta	5–8 kun	Yuqori
32	AI memory (pgvector)	O'rta	15–20 kun	Yuqori
33	AI file attachments	O'rta	8–12 kun	Yuqori
34	Homework AI checker	O'rta	15–20 kun	Juda yuqori
35	AI recommendations engine	O'rta	15–20 kun	Yuqori
36	Calendar integration	O'rta	8–12 kun	O'rta
37	PWA + offline mode	O'rta	1–2 oy	Yuqori
38	Payments (Stripe/Click/Payme)	Kelajak	1–2 oy	Juda yuqori
39	Mobile app (React Native)	Kelajak	3–6 oy	Juda yuqori
40	Marketplace	Kelajak	2–3 oy	Juda yuqori
41	AI Voice input (Whisper)	Kelajak	3–4 hafta	Yuqori
42	Subscription planlar	Kelajak	2–3 hafta	Juda yuqori
43	White label UI (Admin)	Kelajak	1–2 oy	Juda yuqori
44	Multi-tenant architecture	Kelajak	3–6 oy	Juda yuqori
45	Public REST API	Kelajak	2–3 oy	Yuqori
46	SSO Enterprise	Kelajak	1–2 oy	Yuqori
47	AI Tutor Mode	Kelajak	2–3 oy	Juda yuqori
48	Community forum	Kelajak	1–2 oy	O'rta
49	Parent portal	Kelajak	3–4 hafta	O'rta
50	GDPR data export	Kelajak	1–2 hafta	O'rta

7. ROADMAP

Oy 1 — Xavfsizlik va Barqarorlik

Vazifa	Davomiyligi
Role security + email verify + avatars bucket	Hafta 1 (2 kun)
ErrorBoundary + Toast + /reset-password	Hafta 1–2 (3 kun)
Code splitting + profiles.updated_at	Hafta 2 (3 kun)
ProfileContext → barcha sahifalar	Hafta 2–3 (5 kun)
brand.ts + hardcoded tozalash	Hafta 3 (3 kun)
UserAvatar unification	Hafta 3–4 (3 kun)

Oy 2–3 — AI va Core LMS

Vazifa	Davomiyligi
AI rate limiting + token tracking + haqiqiy streaming	2 hafta
AIAssistantPage → feature module	1 hafta
Homework moduli (DB + UI)	2–3 hafta
Notifications + Certificate DB	2 hafta
Leaderboard + Student Dashboard real data	1 hafta
Playwright test suite	2 hafta

Version Belgilari

V1.0 (Oy 3): Xavfsizlik muammolari hal, haqiqiy streaming, homework, notifications, testlar, 80%+ coverage, accessibility AA

V2.0 (Oy 9): AI memory, analytics, payments, marketplace MVP, mobile app, feature-based architecture

V3.0 (Oy 18): AI Tutor, Voice, Multi-tenant, White label, Public API, Enterprise SSO

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